MIL-G-36614A
28 January 1976
SUPERSEDING
MIL-G-36614(DSA-DM)
12 January 1970

MILITARY SPECIFICATION

# GRADUATES, LIQUID, LABORATORY, CORROSION-RESISTING METAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers corrosion-resisting metal graduates, suitable for use in the preparation of pharmaceutical products during field use.

1.2 Classification. Graduates covered by this specification shall be of the following types and sizes, as specified (see 6.1):

Type I - Without handle
Type II - With handle

Size 1 - 500 milliliters (ml.)
Size 2 - 1000 milliliters (ml.)

## 2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636 Boxes, Shipping, Fiberboard.

STANDARDS

Federal

Fed. Std. No. 66 Steel, Chemical Composition and Hardenability.

FSC 6640

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| | |
|---|---|
| Military | |
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Materials. Graduates shall be fabricated of corrosion-resisting steel conforming to AISI 302 or 304 of Fed. Std. No. 66.

3.1.1 Resistance to boil test. Graduates shall show no evidence of corrosion on any exposed surface when tested as specified in 4.4.1.

* 3.2 Design and dimensions. The design and dimensions shall be in strict accordance with Figure 1, for Type I, Size 1 and Figure 2 for Type II, Size 2. As an alternate, Type II, Size 2 shall be graduated in 2-ounce increments. The steel used in the body of the graduates shall have a minimum thickness of 0.022 inches. The thickness of the steel shall be reduced by not more than 0.005 inches as a result of forming, grinding, or polishing operations--except that rounded corners at bottom of the graduates may be reduced to a minimum of 0.010 inches. The steel used in the handle of Type II graduates shall be either rounded in cross section area as indicated in Figure 2, or rectangular in cross section with a minimum thickness of 0.070 inches. The ends of the handle (Type II graduates) shall be either bent and spot welded to the body, or shall be contoured to make contact with the body and then secured to the body by the process of resistance welding.

3.3 Graduation marking. The graduation markings shall be permanently stamped or impressed on the inside of the graduates in the positions shown on the applicable figures. There shall be a double scale, one scale in ounces and the other in milliliters. The ounce scale may be marked in ounces or fluid ounces; the milliliter scale in milliliters or cubic centimeters. Grams may be included with the milliliter scale. Suitable abbreviations may be used for marking.

3.4 Capacity tolerance. The graduation markings shall be accurate to within $\pm$ 3 percent, when tested as specified in 4.4.2.

3.5 Finish. The graduates shall be finished inside and outside with a Number 4 Commercial Finish (AISI) or finer.



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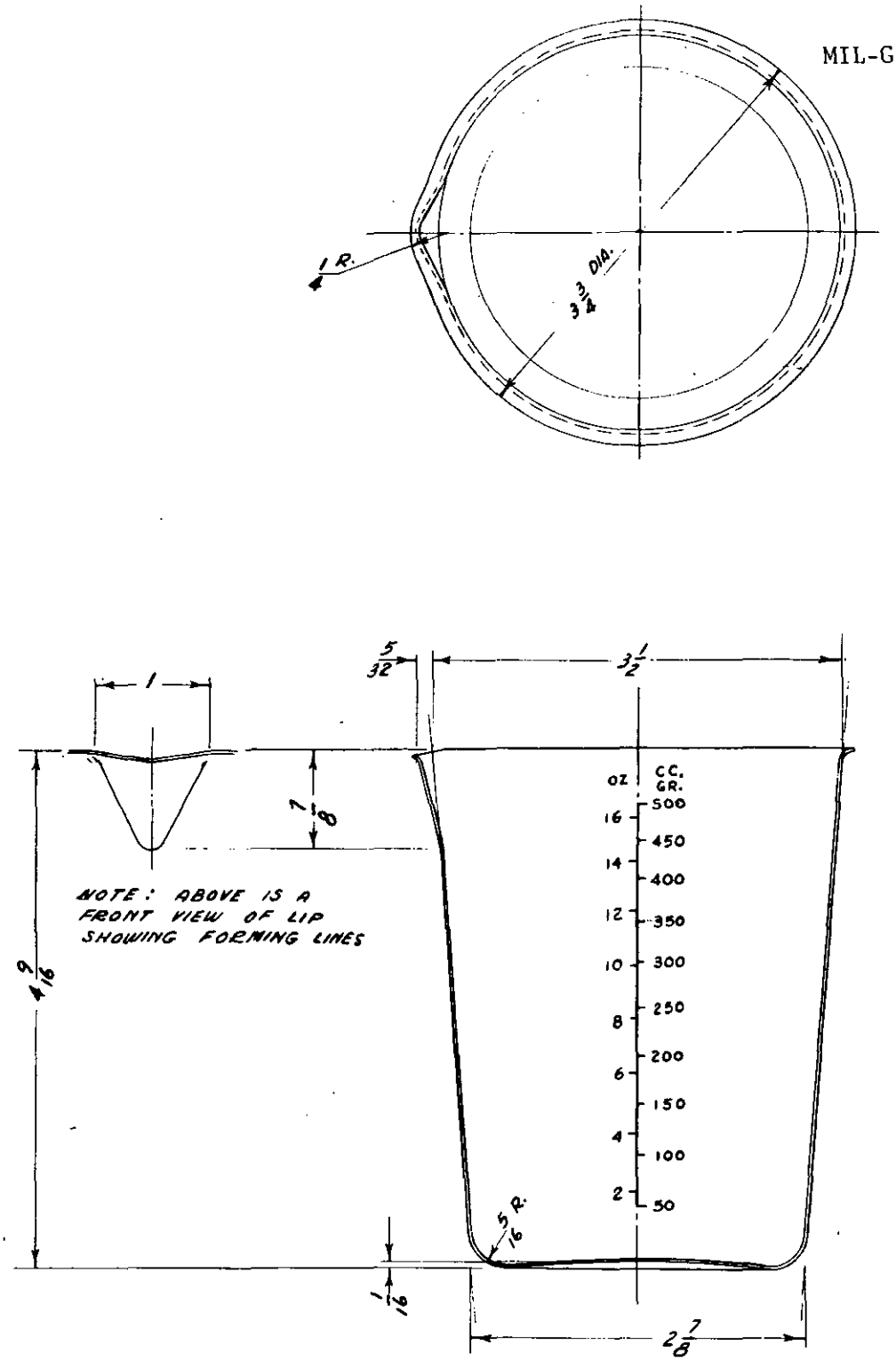
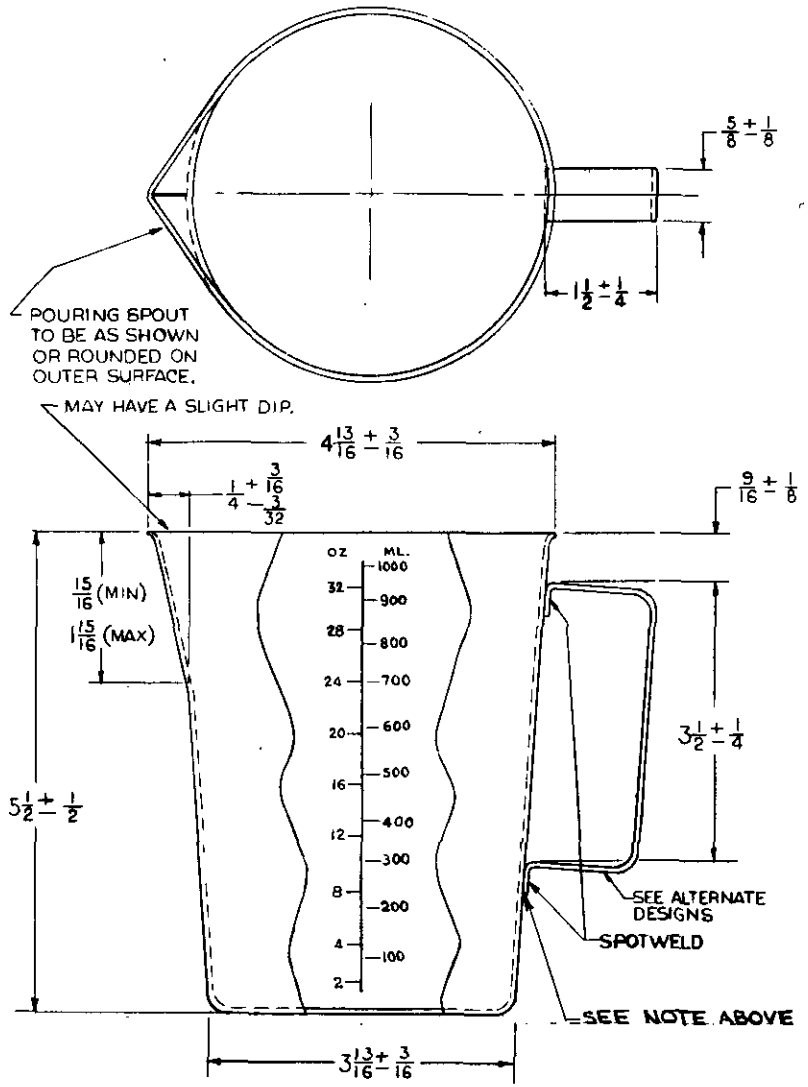


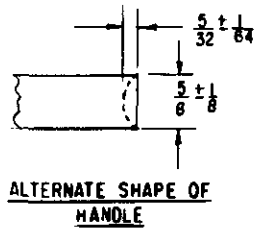

FIGURE 1



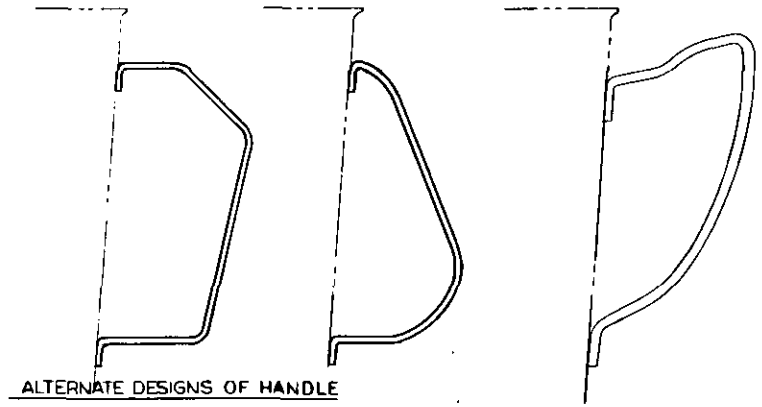
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(AS ALTERNATE IF RESISTANCE WELDING USED TO SECURE HANDLES TO BODY OF GRADUATE, ENDS NOT REQUIRED TO BE BENT AS INDICATED BELOW AND IN ALTERNATE DESIGNS)




ALTERNATE SHAPE OF HANDLE



ALTERNATE DESIGNS OF HANDLE

FIGURE 2

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3.6 Identification marking. Graduates shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.7 Workmanship. Workmanship shall be first class throughout. Graduates shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificate of quality, supplied by the manufacturer of the stainless steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

* 4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable



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quality levels shall be as indicated in Table I. Unit of product for examination purposes shall be the graduate.

Table I. Sampling for examination

| Type defect | Inspection level | AQL (percent defective) |
|---|---|---|
| For visual examination | | |
| Major defects | II | 1.0 |
| Minor defects | II | 2.5 |
| For dimensional examination | S-1 | 2.5 |

4.2.2 Sampling for tests. Sampling for accuracy tests (see 4.4.2) shall be conducted in accordance with MIL-STD-105. Inspection level shall be S-4 and acceptance number shall be zero. Sampling for boil test (see 4.4.1) shall be one graduate, and acceptance number shall be zero.

4.3 Examination. The graduates shall be examined to determine compliance with all requirements contained in this specification. The graduates shall be examined for defects including, but not limited to, those listed in Table II.

Table II. Classification of defects

| Categories | Defects |
|---|---|
| Major | |
| 101 | Graduate not free of sharp edges and not smoothly finished. |
| 102 | Graduate not free of dents. |
| 103 | Handle not secured (Type II) and not smoothly finished. |
| 104 | Graduation markings not complete, correct, legible, or permanent. |
| 105 | Graduate not free of pits and burs. |
| 106 | Graduate not free of scratches. |
| Minor | |
| 201 | Surface behind markings not free of bulging. |
| 202 | Identification marking not complete, correct, legible, or permanent. |

4.3.1 Dimensional examination. The graduates shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.



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* 4.3.2 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

Table III. Examination of preparation for delivery

| Examination | Defect |
|---|---|
| Containers | Not as specified. |
| Quantity | Quantity in interior packages and shipping containers not as specified. |
| Packaging and packing materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted. |
| Markings | Omitted, incorrect, illegible, improper size, location, sequence, or method of application. |

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Boil test. Graduates to be subjected to the boil test shall be tested by the following method: before testing, graduates shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the graduates shall be boiled in distilled water in a glass vessel for 30 minutes and allowed to remain submerged for an additional 24 hours.

4.4.2 Accuracy test. The accuracy of the graduations shall be determined for both the ounce and milliliter scales at approximately 10 percent, 50 percent, and 100 percent capacity, by filling to these capacities with distilled water. Determination of the amount of water may be made by a gravimetric or a volumetric method. For the gravimetric method the



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weighings shall be accurate to the nearest 0.1 gram; for the volumetric method the testing equipment shall be standardized gravimetrically. The balance used for testing and standardizing shall be accurate to the nearest 0.1 gram.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each graduate shall be wrapped in kraft paper of suitable weight or other suitable anti-corrosive material.

5.1.2 Level C. Graduates shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twelve (12) graduates shall be packed in an exterior container constructed in accordance with PPP-B-636, class weather-resistant, grade V3c. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure, waterproofing and strapping shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Twelve (12) graduates shall be packed as specified in 5.2.1, except the box shall be class domestic. Waterproofing and strapping are not required.

5.2.3 Level C. Units of issue shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

## 6. NOTES

6.1 Ordering data. Procurement documents should specify the following:



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a. Title, number and date of this specification.
b. Type and size of graduate required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.3 National Stock Number (NSN) coverage. This specification covers the following items shown in the Federal Supply Catalog:

| NSN | Item Identification |
|---|---|
| 6640-00-427-5025 | GRADUATE, LIQUID, LABORATORY, Corrosion-Resisting Metal, 500 ml., Type I, Size 1 |
| 6640-00-431-4520 | GRADUATE, LIQUID, LABORATORY, Corrosion-Resisting Metal, 1000 ml., Type II, Size 2 |

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0953
CS



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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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